

1 Question (10 points)

Give context-free grammar that generate the following language. The alphabet is $\{0, 1\}$.

$B = \{w \mid \text{the length of } w \text{ is even and its middle symbols are } 00 \text{ or } 11\}$

2 Question (15 points)

Show that the class of context-free languages is closed under the regular operations,

- a. Union
- b. Concatenation
- c. Star

(Hint: Add an extra variable to combine the two CFG's of the languages)

3 Question (10 points)

Give PDA that generate the following language. The alphabet is $\{0, 1\}$.

$B = \{w \mid \text{the length of } w \text{ is even and its middle symbols are } 00 \text{ or } 11\}$

4 Question (15 points)

Are the following languages context-free? Prove your answer.

- $A = \{a^n b^m c^m d^n \mid n, m \geq 0\}$
- $B = \{a^n b^m c^n d^m \mid n, m \geq 0\}$